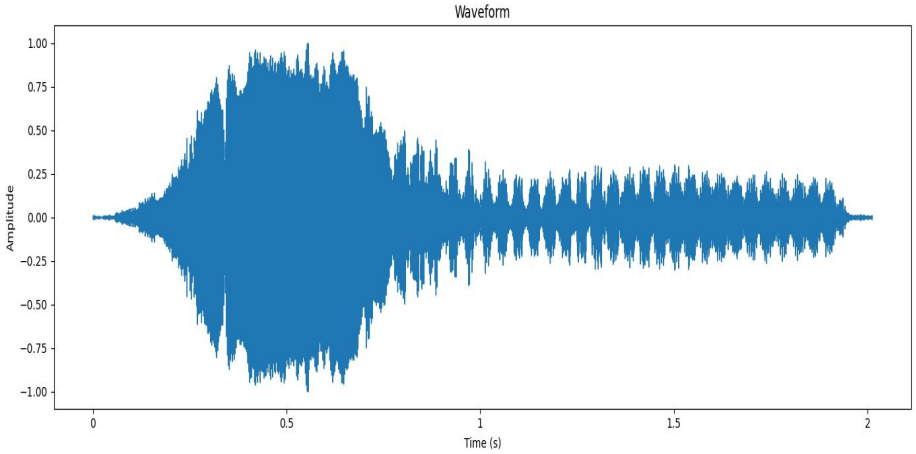
PW SKILLS

Transforming and Visualizing Audio Data

Using Librosa and Torch

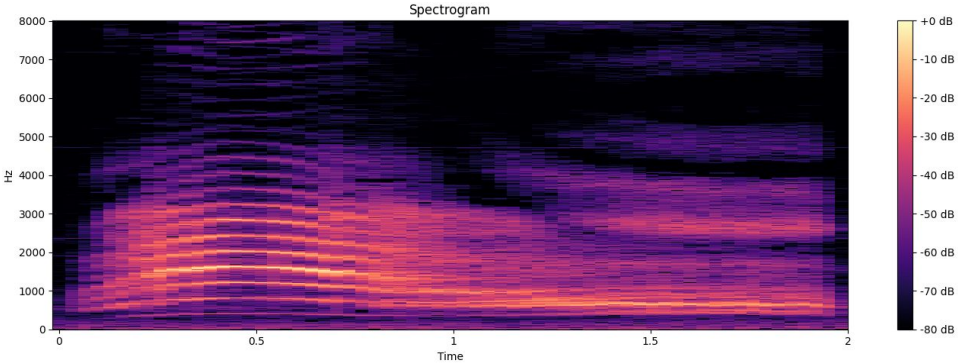
Audio Waveforms

- A graphical representation of how an audio signal **varies over time**.
- Visualize **volume changes** (e.g., loud vs. soft parts).
- Detect **silences, spikes, patterns**, and **noise**.
- Helps in **editing**, **feature extraction**, and **anomaly detection**.
- Waveforms are **basis for more complex features** (like spectrograms).



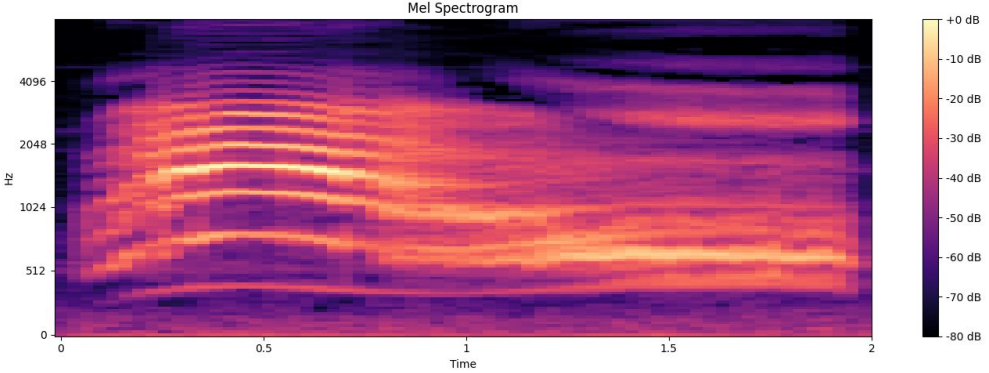
Spectrogram

- A **visual representation** of the **spectrum of frequencies** of a signal as it **varies with time**.
- Show **how energy is distributed** across different frequencies.
- **Bright areas** = Strong frequency components.
- **Dark areas** = Weak or no signal.



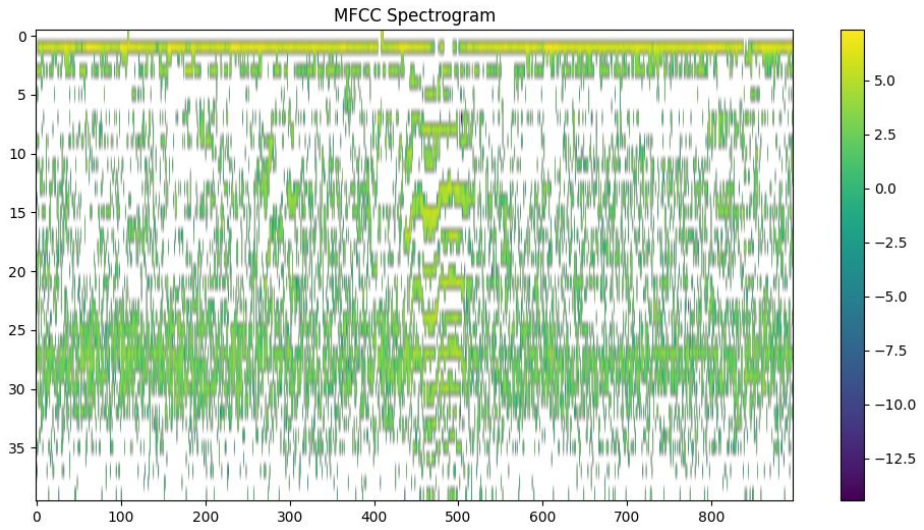
Mel Spectrogram

- A **spectrogram** where the frequency axis is **scaled to the Mel scale**.
- **Mel scale**: A perceptual scale that **mimics how humans hear** (better resolution at lower frequencies, compressed at higher frequencies).
- Human ears **don't perceive frequencies linearly**.
- Mel spectrograms focus **more detail** where it **matters to human perception**.



MFCC

- It's a **compact, information-rich sequence** of vectors (coefficients) over time.
- You can plot MFCCs over time — **looks like a spectrogram**, but each "frequency axis" now is **coefficient index**, not physical frequency.
- They capture the **overall shape** of the human auditory spectrum.
- Extremely popular for **speech recognition**, **speaker identification**, **emotion recognition**, etc.
- They remove unnecessary details but **keep the essence** of human hearing.



What is Librosa?

- A Python package for **music and audio analysis**.
- Built on top of **NumPy**, **SciPy**, and **scikit-learn**.
- Focused on **feature extraction**, **visualization**, and **audio processing**.

Thanks

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